HEALTHCARE SUPPLY CHAIN ANALYTICS | WEEK 2

Data Integrity Report

Task 11: SQL Staging, Duplicate and Key Integrity Checks

Item	Details
Project	Healthcare Supply Chain Analytics
Task owner	Hossam, Technical Analytics Lead
Reporting period	Sunday 18 October 2026 to Thursday 22 October 2026
Report date	19 October 2026
Database	SupplyChainProject_DB (SQL Server Express), staging layer
Source data	Data_Final_project_IR.xlsx
Script	Task11_Integrity_Checks_FINAL.sql (read-only, re-runnable)

1. Executive Summary

The staging layer was built from the project workbook and checked for key uniqueness, duplicates, referential integrity, invalid item codes and entity classification. All reference counts from the Week 2 plan were reproduced exactly, with zero variance. No raw or staging row was modified or deleted during the checks.

248	5	37	65
Full-duplicate rows (162 + 19 + 67)	Purchase order keys with conflicting quantity	Orphan rows for hospital H-E (23 + 4 + 10)	Invalid item-code rows (I099, I000, I3O1)

Key findings

- The shipments table (in) is clean on its key: 46 rows and 46 distinct ids.
- inventory_daily holds 162 full-duplicate rows, purchase_orders holds 19 full duplicates plus 5 po_id groups whose qty_ordered values conflict, and patient_cases holds 67 full duplicates.
- Hospital H-E does not exist in the hospital reference (H-A to H-D) and appears in all three child tables. It is flagged as an orphan and is not merged into any other hospital.
- Three invalid item codes were found: I099 and I000 (outside I001 to I030) and I3O1 (letter O instead of zero). They are flagged and not corrected by guessing.
- Supplier S09 is the Regional Distribution Center (RDC), one purchase order row, and must be excluded from vendor KPIs.

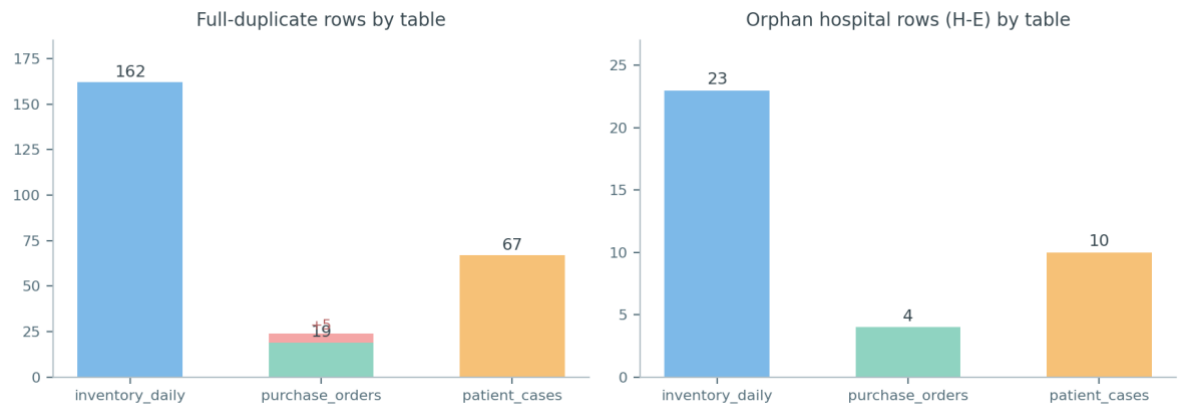


Figure 1. Full-duplicate rows and H-E orphan rows by table.

2. Scope and Method

The checks follow the governing principle of Week 2: raw data is never edited. Each staging table is reached through a synonym, and every result in this report comes from a SELECT query, so the script can be re-run at any time with identical output.

Staging table	Source sheet	Rows in source
Staging_in	in (shipments)	46
Staging_inventory_daily	inventory_daily	10,793
Staging_purchase_orders	purchase_orders	2,187
Staging_patient_cases	patient_cases	5,665
Staging_hospitals / items / suppliers	hospitals / items / suppliers	4 / 30 / 9

Section 1 of the script returns the loaded row count of each staging table, to be compared with the source counts above before cleaning starts.

3. Key Uniqueness and Duplicates

Duplicates were tested at full-row level, not on the key alone, so that identical copies and conflicting copies are separated.

Table	Key	Rows	Distinct keys	Full duplicates	Conflicting groups	Expected rows after cleaning
in	id	46	46	0	0	46
inventory_daily	record_id	10,793	10,631	162	0	10,631
purchase_orders	po_id	2,187	2,163	19	5	2,163
patient_cases	case_id	5,665	5,598	67	0	5,598

For purchase_orders, 24 rows share a key with another row: 19 are identical copies and 5 are the second row of a group whose qty_ordered differs. The 5 conflicting groups are listed below.

po_id	Item	Hospital	qty_ordered values	Note
PO-000025	I002	H-A	415 and 425	Status: Partially Received
PO-000160	I009	H-A	195 and 205	Status: Received
PO-000358	I018	h-a	260 and 270	Hospital id needs normalization
PO-000503	I024	H-A	305 and 315	Status: Received
PO-001388	I006	H-C	42 and 52	Status: Received

Proposed treatment: keep the first row under a fixed, declared order, log the conflicting row, and ask the Data Quality lead to sign off on the ordering basis. Nothing is deleted at this stage.

4. Referential Integrity: hospital_id

The approved reference contains four hospitals: H-A, H-B, H-C and H-D. Raw hospital_id values appear in 17 spellings per table (for example h-a, H_A and trailing spaces). After safe normalization only (trim, upper case, underscore to hyphen), a single value remains outside the reference.

Child table	Rows with non-standard spelling	Orphan value	Orphan rows
inventory_daily	610	H-E	23
purchase_orders	117	H-E	4
patient_cases	312	H-E	10
Total	**1,039**		**37**

H-E is kept in the data, flagged as Orphan, and not merged with any hospital. In the SQL run the raw check returned 27 rows (9 distinct values in each of the 3 tables, because SQL Server ignores letter case and trailing spaces in grouping) and the normalized check returned 3 rows, one H-E group per table.

5. Domain Check: Item Codes

Valid item codes run from I001 to I030. Codes outside the range or with a letter in place of a digit are flagged without correction. Codes that are valid but written in lower case or with spaces are a formatting issue handled in the cleaning layer.

Table	I099	I000	I3O1	Case or space only	Invalid rows
inventory_daily	16	7	12	180	35
purchase_orders	4	0	4	0	8
patient_cases (key_drug_item_id)	0	22	0	50	22
Total	**20**	**29**	**16**	**230**	**65**

6. Entity Classification: Supplier S09

Supplier S09 appears once in purchase_orders (PO-002258) under the name SCMS - Regional Distribution Center (RDC). It is an internal distribution center, not a commercial vendor, and is classified as RDC so that it is excluded from vendor performance KPIs such as on-time delivery and fill rate.

7. Missing and Invalid Values

Profiling of each table found the following issues. They are reported here for the cleaning team and are not corrected in staging. Blank values stay NULL and are never converted to zero.

Column	Finding	Rows
Shipments (in)		
country	Encoding error in Cote d'Ivoire (CÃ´te d'Ivoire)	6
pq first sent to client date	Text "Pre-PQ Process" inside a date column	21
po sent to vendor date	Text "Date Not Captured" inside a date column	1
weight (kilograms)	Text such as "See ASN..." and "Weight Captured Separately"	8
freight cost (usd)	Text such as "See ASN..." and "Freight Included in Commodity Cost"	8
Other columns	Blank: dosage 8, receiving_inspection_result 6, batch_number 3, expiry_date 2	19
Inventory (inventory_daily)		
date	Mixed formats: dd/mm/yyyy 417, dd-Mon-yy 108, Excel serial number 43	568
closing_stock	Text values: "-" 15, "?" 13, numbers with thousands separators 15	43
stock_value_egp	Text with "EGP" prefix	86
stockout_flag	Mixed Y / N / TRUE / FALSE instead of 0 / 1	203
received_qty	Negative quantities (flag Negative_Received)	43
date	Outside 1 Jan to 31 Mar 2026 (flag Out_of_Period)	43
Blank values	unmet_demand_qty 1,678; days_of_cover 624; dispensed_qty 348; reorder_point 255; avg_daily_dispensed_7d 244; stock_value_egp 236; closing_stock 154; received_qty 99; requested_qty 99	-

Column	Finding	Rows
Purchase orders (purchase_orders)		
po_status	14 raw spellings for 4 standard statuses	211
unit_cost_egp	Dollar format such as $0.54 (flag Currency_Mismatch)	23
unit_cost_egp	Text with "EGP" prefix	39
qty_received	Text: "unknown" 8, "see GRN" 7; blank 179	15
order_date	Mixed formats: dd/mm/yyyy 61, dd-Mon-yy 23, Excel serial number 8	92
received_date	Blank (may be legitimate for orders not yet received)	194
Patient cases (patient_cases)		
admission_date	Mixed formats: dd/mm/yyyy 223, dd-Mon-yy 50, Excel serial number 13	286
age_years	Text in days (for example "12 days"); blank 207	44
gender	Variants f, m, Female, female, Male, male; blank 133	518
severity	Numeric codes 1, 2, 3 instead of Low, Moderate, High; blank 247	107
icu_admission	Variants n, No, y, Yes instead of N / Y	238
treatment_delay_hours	Text with "h" suffix (for example "0.4h"); blank 103	59
length_of_stay_days	Text with "days" suffix; blank 411	52
Blank values	discharge_date 449; outcome 39	-

Slash dates must be converted with the explicit format dd/mm/yyyy only, never by automatic guessing.

8. Exceptions Register

Group-level list of exceptions with example keys. The row-level lists are produced by the queries in the script.

Table	Example keys	Raw value	Issue type	Rows	Proposed action
purchase_orders	PO-000025, PO-000160, PO-000358, PO-000503, PO-001388	Different qty_ordered for the same po_id	Conflicting key	5 groups	Keep first row in fixed order; log the other; quality sign-off
inventory_daily	record_id repeated	Identical copies	Full duplicate	162	Remove in clean layer only; log keys
purchase_orders	po_id repeated	Identical copies	Full duplicate	19	Remove in clean layer only; log keys
patient_cases	case_id repeated	Identical copies	Full duplicate	67	Remove in clean layer only; log keys
inventory_daily	1730, 1792	H-E	Orphan hospital	23	Keep and flag Orphan; do not merge
purchase_orders	PO-000572, PO-001085	H-E	Orphan hospital	4	Keep and flag Orphan; do not merge
patient_cases	C000185, C000254	H-E	Orphan hospital	10	Keep and flag Orphan; do not merge
inventory_daily	62, 1272	I099	Out-of-range item	16	Flag Invalid_Item; no guessing
inventory_daily	206, 767	I000	Out-of-range item	7	Flag Invalid_Item; no guessing
inventory_daily	340, 903	I3O1	Malformed item	12	Flag Invalid_Item; no guessing
purchase_orders	PO-000551, PO-000611	I099	Out-of-range item	4	Flag Invalid_Item; no guessing
purchase_orders	PO-000482, PO-000834	I3O1	Malformed item	4	Flag Invalid_Item; no guessing
patient_cases	C000134, C000639	I000	Out-of-range item	22	Flag Invalid_Item; no guessing
purchase_orders	PO-002258	S09	RDC, not a vendor	1	Classify as RDC; exclude from vendor KPIs
purchase_orders	PO-000079, PO-000133	$ price format	Currency mismatch	23	Flag Currency_Mismatch; no conversion
inventory_daily	68, 313	Negative received_qty	Negative value	43	Flag Negative_Received; keep value

9. Reconciliation with Reference Counts

Check	Reference	Actual	Variance	Status
in: COUNT(*) = COUNT(DISTINCT id)	46 = 46	46 = 46	0	Pass
inventory_daily full duplicates	162	162	0	Pass
purchase_orders full duplicates	19	19	0	Pass
purchase_orders conflicting po_id groups	5	5	0	Pass
patient_cases full duplicates	67	67	0	Pass
H-E orphan rows (inventory / PO / cases)	23 / 4 / 10	23 / 4 / 10	0	Pass
Item code examples I3O1, I099, I000	Present	Present	0	Pass
S09 classified as RDC	Yes	Yes (1 row)	0	Pass
Negative received_qty (for Task 12-13)	43	43	0	Pass
Out-of-period inventory rows (for Task 12-13)	43	43	0	Pass

Query outputs seen in SQL Server Management Studio match these figures: 162 duplicated record_id values in inventory, 5 conflicting purchase order groups and 67 duplicated case_id values.

10. Acceptance Criteria

Criterion	Evidence	Status
Re-runnable staging script	Task11_Integrity_Checks_FINAL.sql	Met
Integrity Report with counts, key uniqueness, duplicates, orphans, invalid codes and missing values	This report, sections 3 to 7	Met
Exceptions list with table, key, raw value, issue and proposed action	Section 8	Met
Raw not modified and no row deleted from staging	Script contains SELECT statements only	Met
Every number tied to a reviewable query	Script sections 1 to 8	Met

Check to script section

Check	Script section	Result used in this report
Staging row counts	1	Rows per staging table
Primary key uniqueness: in	2	Total rows, distinct ids, duplicate count
Full duplicates: inventory_daily	3	Full-duplicate rows and repeated record_id values
Duplicates and conflicts: purchase_orders	4	Full-duplicate rows and conflicting po_id groups
Full duplicates: patient_cases	5	Full-duplicate rows and repeated case_id values
Hospital referential integrity	6	Raw spellings and normalized orphans
Item code range	7	Out-of-range and malformed codes

Check	Script section	Result used in this report
Supplier S09	8	RDC classification

11. Handoff to Next Tasks

- Tasks 12 and 13 (Data Preparation): build the clean layer, remove the 248 full-duplicate rows from the clean copy only, normalize hospital_id and po_status, and add the flags named in this report (Invalid_Item, Orphan, Out_of_Period, Negative_Received, Currency_Mismatch).
- Task 14 (Data Quality review): approve the ordering basis for the 5 conflicting purchase orders, the mapping for item code formats, and the treatment of H-E.
- Task 15 (Reconciliation): use the expected clean row counts in section 3 as the after-cleaning targets: 10,631 inventory rows, 2,163 purchase orders and 5,598 patient cases.